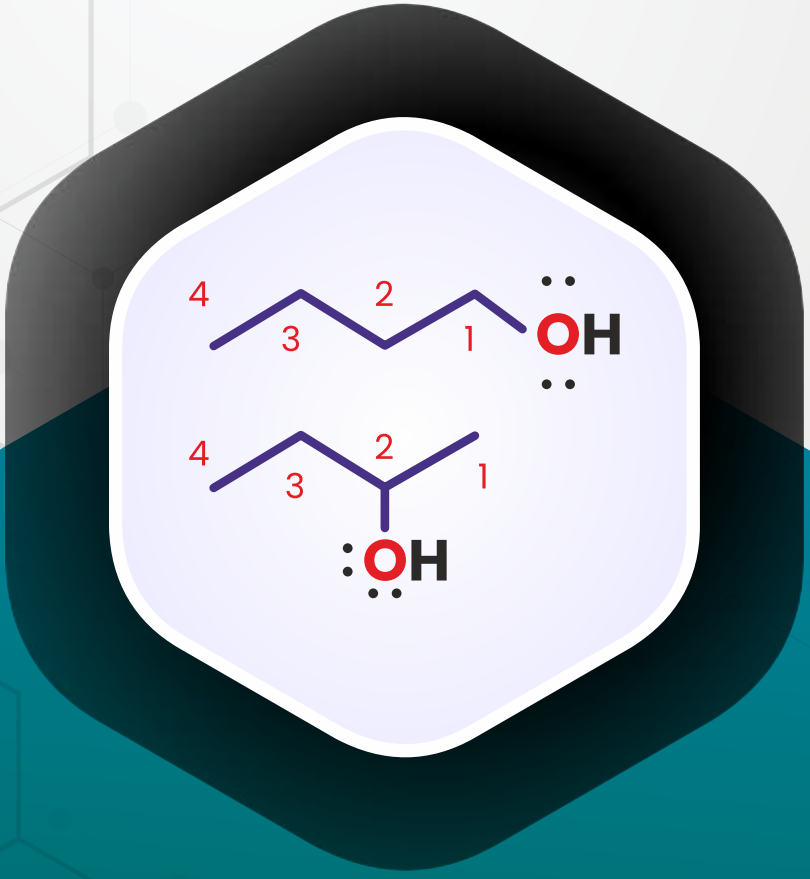
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Nomenclature & Common Names

(English Medium)

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-ESSENTIAL COMMON NAMES-

ALKANE

1 $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH-CH_3$ Isopentane

ALKENE

2 $CH_2=C=CH_2$ Allene

ALKYL HALIDE

3 $\underset{\displaystyle CHCl_2}{\underset{|}{CHCl_2}}$ Westron (Solvent)

4 $ClCH=CCl_2$ Westrosol or Triclean (Solvent)

ALCOHOL

5 $\underset{\displaystyle CH_2-OH}{\underset{|}{CH_2-OH}}$ Glycol or Ethylene Glycol

6 $\underset{\displaystyle OH}{\underset{|}{CH_2}}-\underset{\displaystyle OH}{\underset{|}{CH}}-\underset{\displaystyle OH}{\underset{|}{CH_2}}$ Glycerol

7 $CH_2=CH-CH_2-OH$ Allyl Alcohol

8 $CH_2=CH-OH$ Vinyl Alcohol

ETHER

9 $C_6H_5-O-CH_3$ Anisole (Methyl Phenyl Ether)

KETONE

10 CH_3COCH_3 Acetone

CARBOXYLIC ACID

11 $HO-\underset{\displaystyle CH_2-COOH}{\underset{|}{CH}}-COOH$ Malic acid

12 $HO\ CH_2\ COOH$ Glycolic Acid

13 $CH_2\begin{matrix}COOH\\COOH\end{matrix}$ Malonic acid

14 $\underset{\displaystyle CH_2-COOH}{\underset{|}{CH_2-COOH}}$ Succinic acid

15 $\underset{\displaystyle CH_2-COOH}{\underset{|}{\underset{|}{CH_2}}}$ Glutaric acid

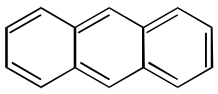
16 $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_2-\overset{\displaystyle O}{\overset{\|}{C}}-O-C_2H_5$ Aceto Acetic Ester **(AAE)** or Ethyl Aceto Acetate

N–DERIVATIVES

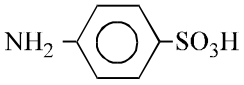
17 $CH_2=CH-C\equiv N$ Vinyl Cyanide or Acrylo Nitrile

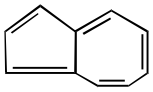
18 $NH_2-\overset{\displaystyle NH}{\overset{\|}{C}}-NH_2$ Guanidine

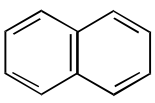
AROMATIC COMPOUNDS

19  Anthracene

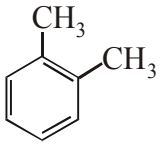
20  Thiophene

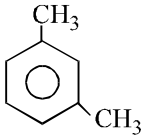
21  Sulphanilic acid

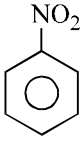
22  Azulene

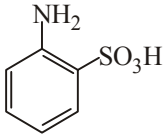
23  Napthalene

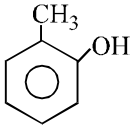
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

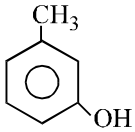
24  o–xylene

25  m-xylene

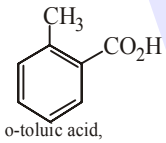
26  Nitrobenzene (oil of mirbane)

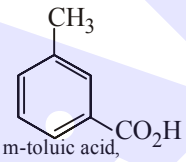
27  Orthanilic Acid

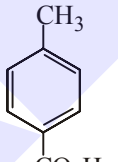
28  o-Cresol

29  m-Cresol

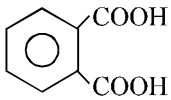
30 $C_6H_5CO_3H$ Perbenzoic acid

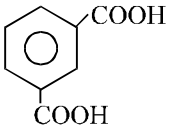
31 
o-toluic acid, m.p. 105°C

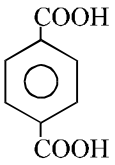

m-toluic acid, m.p. 111°C

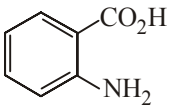

p-toluic acid, m.p. 180°C

Toluic acids

32  Phthalic acid

33  Isophthalic acid

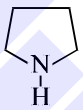
34  Terephthalic acid

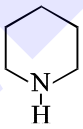
35 

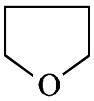
Anthranilic acid (o-aminobenzoic acid)

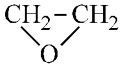
36 C_6H_5CHO Benzaldehyde

HETROCYCLIC COMPOUNDS

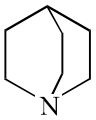
37  Pyrrolidine

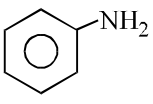
38  Piperidine

39  Tetrahydrofuran (THF)

40  Oxirane or Ethylene Oxide or

Oxo Cyclo Propane

41  Quinuclidine

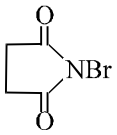
42  Aniline

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

SOME REAGENTS

43 Grignard's reagent RMgX

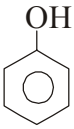
44 NBS N-Bromosuccinimide



POLAR PROTIC SOLVENTS

45 H –O–H Water

46 R–O–H Alcohol

47  Phenol

48 $CH_3–\underset{\underset{O}{\|}}{C}–OH$ Acetic acid

49 HF Hydrogen Fluoride

50 NH_3 Ammonia

POLAR APROTIC SOLVENTS

51 DMS Dimethyl sulphide $CH_3–S–CH_3$

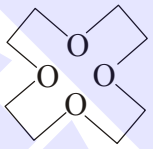
52 DMSO Dimethyl sulphoxide $Me_2S=O$

53 HMPT Hexamethylphosphoramide
or
HMPTA $O=P–(NMe_2)_3$

54 DMF Dimethyl formamide

$H–\underset{\underset{O}{\|}}{C}–NMe_2$

55 Crown ethers Cyclic polyethers

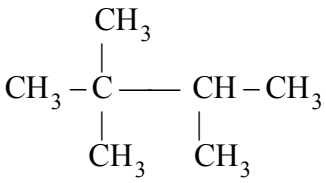


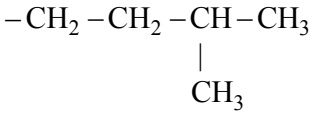
(12 – C – 4)

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

-DESIRABLE COMMON NAMES-

ALKANES

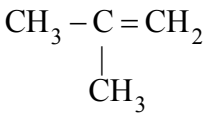
1  Triptane

2  Isopentyl Group

ALKENES

3 $CH_3–CH_2–CH=CH_2$ α–Butylene

4 $CH_3–CH=CH–CH_3$ β–Butylene

5  Iso Butylene

ALKYNES

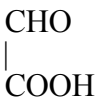
6 $HC\equiv CH$ Purified Acetylene or Norcelyne

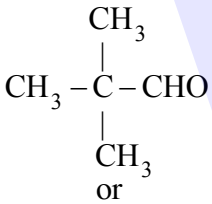
7 $CH_3–C\equiv CH$ Allylene

ETHER

8 $CH_3CH(OCH_3)_2$ Methylal

ALDEHYDE

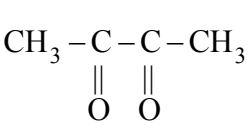
9  Glyoxalic acid

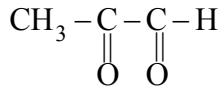
10  Pivaldehyde

or

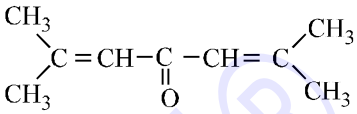
$(CH_3)_3C–CHO$

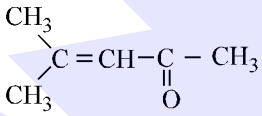
11 $(CH_3)_2CHCHO$ Isobutyraldehyde

12  Dimethyl Glyoxal

13  Methyl Glyoxal or Pyruvialdehyde

KETONE

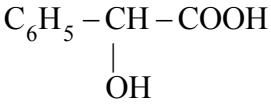
14  Phorone

15  Mesityl Oxide

16 $H_2C=C=O$ Ketene

CARBOXYLIC ACID

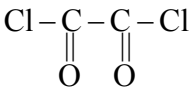
17 $CH_3–CO–COOH$ Pyruvic Acid

18  Mendalic Acid

19 NH_2COOH Carbamic Acid (Amino formic Acid)

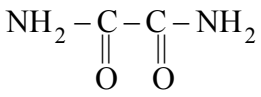
20  Oxalic acid

ACID DERIVATIVES

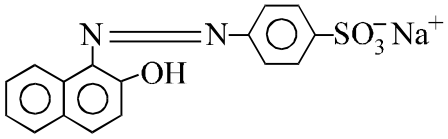
21  Oxalyl Chloride

22 NH_2COONH_4 Ammonium Carbamate

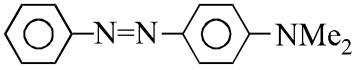
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3.1Nomen-EDE-A & B.Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

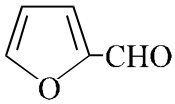
23  Oxanamide

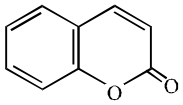
AROMATIC COMPOUNDS

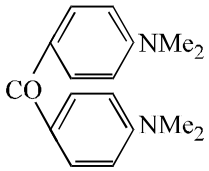
24 

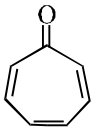
Orange II

25  Butter Yellow

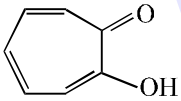
26  Furfural

27  Coumarine

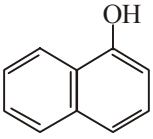
28  Michler's Ketone

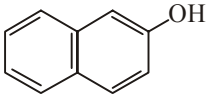
29  Tropone

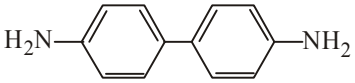
(Cycloheptatrienone)

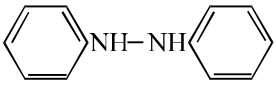
30  Tropolone

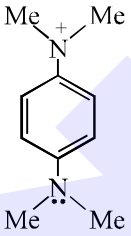
(Cycloheptatrienolone)

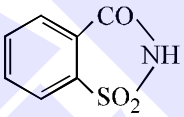
31  α–naphthol

32  β–naphthol

33  Benzidine

34  Hydrazobenzene

35  Wurster salts

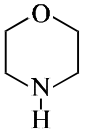
36 

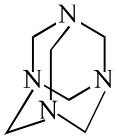
Saccharin (o-sulphobenzoic imide)

37 $C_6H_5COCOC_6H_5$ Benzil

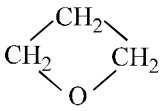
38 $(C_6H_5)_2C(OH)CO_2H$ Benzilic acid

HETROCYCLIC COMPOUNDS

39  Morpholine

40 

Hexa-methylenetetramine or Urotropine

41  Oxitane

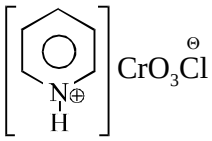
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

SOME REAGENTS

43 LAH — Lithium aluminium hydride : $LiAlH_4$

44 SBH — Sodium borohydride $NaBH_4$

45 PCC — Pyridinium chlorochromate

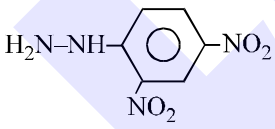


46 Raney Nickel — Ni–Al alloy

47 Wilkinson's catalyst — Tris(Triphenylphosphine) chlororhodium (I) $(PPh_3)_3RH^{\oplus}Cl^{\ominus}$

48 Bayer's reagent — 1% dil. alkaline aq.sol. of $KMnO_4$

49 Braddy's reagent 2,4 DNP



50 Liemieux reagent — $NaIO_4$ + dil.alk.$KMnO_4$

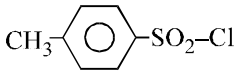
51 TEL — Tetra ethyl lead

52 Gillman's reagent — $R_2CuLi/[R_2Cu]^- Li^+$

53 Tollen's reagent — alk. sol. of $AgNO_3$

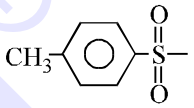
54 Fehling's reagent — alk. sol. of $CuSO_4$

55 Hinsberg's reagent

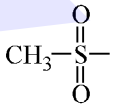


SOME GROUPS

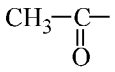
56 Ts — Tosyl



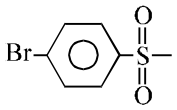
57 Ms — Mesyl



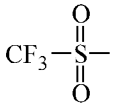
58 Ac — Acyl



59 Bs — Brosyl



60 Tf — Triflate



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